Modern HA-ISP CollapseCore — Network Overview

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Summary

Modern HA-ISP CollapseCore is a Packet Tracer enterprise campus with no single point of failure from the user port to the Internet. It is built on a collapsed core, two independent firewalls and two separate ISPs, and every layer has a working second path.

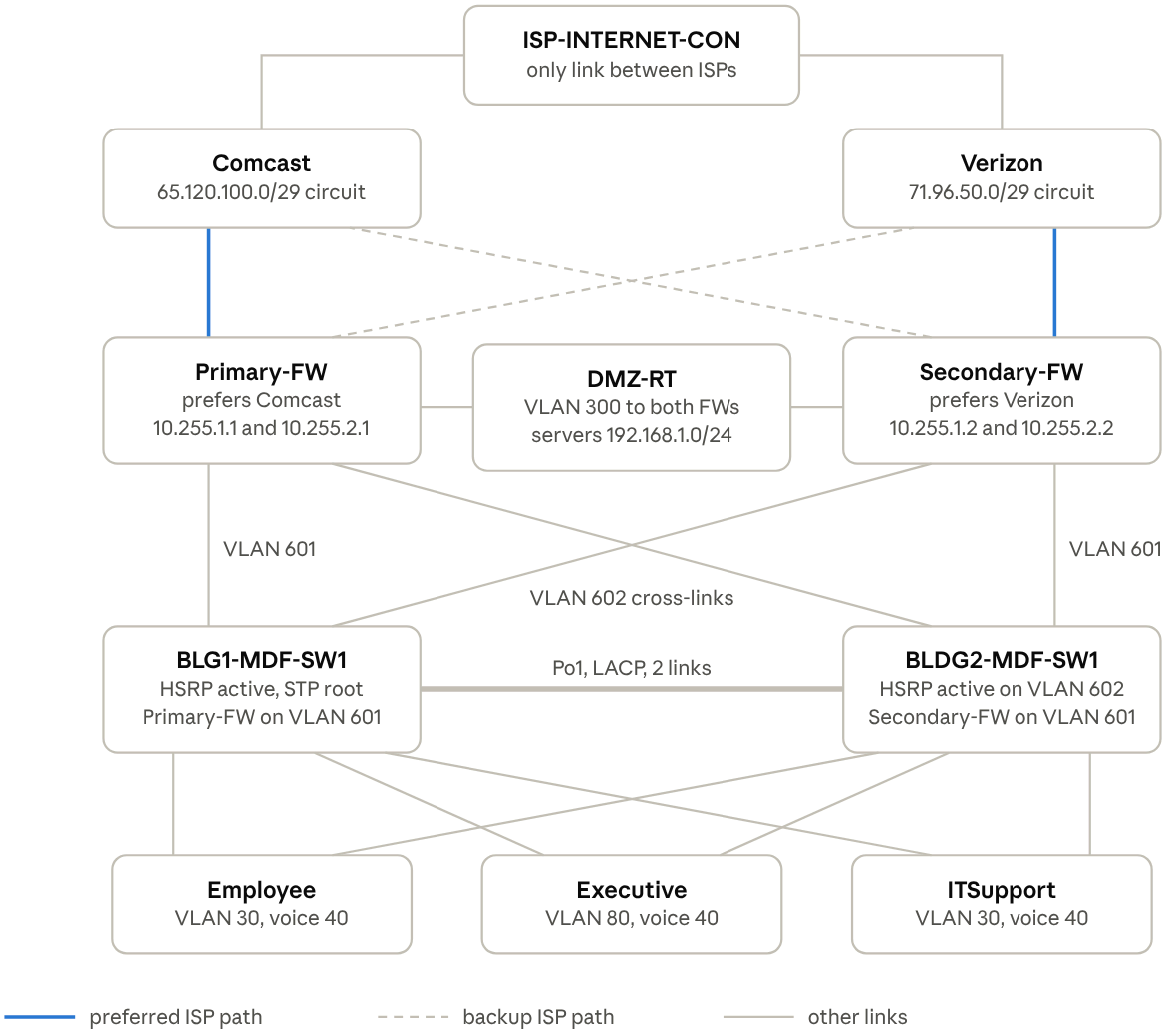
- **Campus:** two Layer 3 cores (BLG1-MDF-SW1 and BLDG2-MDF-SW1) joined by a 2-link LACP channel, with three dual-homed access switches (Employee, Executive, ITSupport).
- **Gateways:** HSRPv2 on every VLAN, with BLDG1 as primary and BLDG2 as backup. VLAN 602 reverses this so each core carries one firewall path.
- **Perimeter:** two Cisco ASA 5506 firewalls (Primary-FW and Secondary-FW), each connected to both cores, both ISPs and the DMZ router.
- **Internet:** simulated Comcast and Verizon networks, kept separate and joined only through an upstream router, as two real ISPs would be.
- **Services:** an eight-server DMZ (DHCP, DNS, web, email, RADIUS, NTP, TFTP), Cisco CME voice, and a WLC with three lightweight APs.
- **Management:** SSH only, RADIUS login with local fallback, NTP with authentication, and port security on every access port.

The lab was built as a CCNP-oriented portfolio piece. It demonstrates campus design, first-hop redundancy, firewall policy and NAT, dual-ISP routing, wireless, voice and AAA in one working network.

Topology and physical layout

The network has five layers, and each one is doubled: ISPs, firewalls, cores, access switches and the DMZ path. The diagram shows the intended paths.

Every layer has a second path, from user port to ISP



topology · ISPs, firewalls, cores, access switches

Primary-FW and Secondary-FW each prefer a different ISP and cross-connect to both cores, so any one device can fail without cutting users off.

Key cabling:

- Each firewall connects to both ISP switches (Gi1/4 to Comcast, Gi1/5 to Verizon), to both cores (Gi1/1 and Gi1/2) and to DMZ-RT (Gi1/3).
- Each ISP circuit passes through a cloud and modem pair to its ISP router.
- Each access switch has one uplink to each core.
- The WLC connects to BLDG1 Gi1/0/24. CME-RT connects to BLDG1 Gi1/0/6 and BLDG2 Gi1/0/6.
- DMZ-SW1 uplinks to DMZ-RT Fa0/0 and holds all eight DMZ servers plus a server PC.

Addressing plan

The campus uses 128.137.232.0/22 split into five VLAN subnets. Firewall and DMZ transit use 10.255.0.0/16. The full plan, with masks, ranges and every static address, is in the Excel workbook delivered with this doc.

Network	CIDR	VLAN	Gateway (HSRP VIP)	Purpose
128.137.232.0	/24	30 CLIENT	128.137.232.254	Employees, IT, wireless clients
128.137.233.0	/24	40 VOICE	128.137.233.254	IP phones and CME (.252)
128.137.234.0	/27	70 PRINTER	128.137.234.30	Printers
128.137.234.32	/27	80 EXECUTIVE	128.137.234.62	Executive users, wireless
128.137.234.64	/27	999 MGMT	128.137.234.94	Switch, core and WLC management
128.137.234.96	/27	100 GUEST	128.137.234.126	Guest wireless
10.255.1.0	/29	601 FW-TRANSIT-A	10.255.1.5	Cores to firewalls, path A
10.255.2.0	/29	602 FW-TRANSIT-B	10.255.2.5	Cores to firewalls, path B
10.255.30.0	/29	300	none	Firewalls to DMZ-RT
192.168.1.0	/24	200	192.168.1.1 (DMZ-RT)	DMZ servers
65.120.100.0	/29	10	65.120.100.1	Comcast circuit
71.96.50.0	/29	20	71.96.50.1	Verizon circuit
75.75.75.0	/24	none	75.75.75.1	Comcast DNS (.75) and web (.74)
68.237.161.0	/24	none	68.237.161.1	Verizon DNS (.14) and web (.15)
203.0.113.0	/30	none	none	Comcast-L3 to INTERNET-RT

Network	CIDR	VLAN	Gateway (HSRP VIP)	Purpose
203.0.113.4	/30	none	none	Verizon-L3 to INTERNET-RT

VLAN 90 is the native VLAN on every inter-switch trunk and carries no hosts or IP subnet.

Static management addresses: access switches Employee .71, Executive .72 and ITSupport .73 on VLAN 999, the WLC at 128.137.234.70, and DMZ-SW1 at 192.168.1.10. Each core has its own address in every user subnet, with BLDG1 on the lower one and BLDG2 on the higher one (128.137.232.1 and .2). On the firewall transit subnets Primary-FW and Secondary-FW take .1 and .2, and the cores take .3 and .4.

Layer 2 design

BLDG1 is the spanning-tree root and BLDG2 the backup root for every VLAN, so traffic always has one predictable path.

- **Spanning tree:** rapid-PVST on all switches. Root priority is 4096 on BLDG1 and 8192 on BLDG2 for VLANs 30, 40, 70, 80, 90, 100, 601, 602 and 999.
- **Core-to-core link:** Port-channel1 is a 2-link LACP bundle (Gi1/1/1 and Gi1/1/2 on each core, single-mode fiber). It trunks the user, management and firewall-transit VLANs with native VLAN 90.
- **Access uplinks:** each access switch has Fa0/1 to BLDG1 and Fa0/2 to BLDG2 as trunks. Employee uses Gi1/0/3 on each core, Executive Gi1/0/4 and ITSupport Gi1/0/5. Allowed VLANs are 30, 40, 70, 80, 90, 100 and 999. Spanning tree blocks the BLDG2 side until BLDG1 fails.
- **Firewall ports:** Gi1/0/1 is an access port in VLAN 601 and Gi1/0/2 in VLAN 602. Primary-FW uses the BLDG1 port for transit A and the BLDG2 port for transit B. Secondary-FW is cabled the opposite way.
- **User ports:** Fa0/3 to Fa0/19 are VLAN 30 (Employee, ITSupport) or VLAN 80 (Executive) with voice VLAN 40. Each has sticky port security (max 5, violation restrict), PortFast and BPDU guard. Fa0/20 is the printer port in VLAN 70.
- **Wireless ports:** Fa0/21 on each access switch is an AP trunk with native VLAN 999 and VLANs 30, 80 and 100. The WLC connects to BLDG1 Gi1/0/24 with the same VLAN set.
- **DMZ:** DMZ-SW1 puts all servers in VLAN 200 with sticky port security (max 2) and gives them one uplink to DMZ-RT.

Layer 3 design

HSRP gives hosts a gateway that survives a core failure, and OSPF area 0 carries routes between the cores, firewalls and DMZ router.

- **HSRP:** version 2 on every SVI with preempt. BLDG1 has priority 105 on VLANs 30, 40, 70, 80, 100, 601 and 999. BLDG2 has priority 105 on VLAN 602, so each core is active for one firewall transit path.
- **Virtual gateways:** the .254, .30, .62, .126 and .94 addresses in the addressing plan, plus 10.255.1.5 and 10.255.2.5 on the firewall transit VLANs.
- **DHCP relay:** every user VLAN and VLAN 999 relays to the DHCP server at 192.168.1.2.
- **OSPF area 0:** router IDs are 10.255.255.11 (BLDG1), .12 (BLDG2), .1 (Primary-FW), .2 (Secondary-FW) and 10.255.255.30 (DMZ-RT). Cores and DMZ-RT run passive-interface default and unpassive only their transit VLANs.
- **Default route:** only Primary-FW originates a default into OSPF. Secondary-FW does not, so the cores never see two equal-cost defaults.
- **Path preference:** Vlan602 has OSPF cost 50 on both cores and Primary-FW Gi1/2 has cost 50, which keeps return traffic on one path. Secondary-FW interfaces carry cost 100, which makes its own paths toward campus and DMZ networks less attractive than Primary-FW's.
- **Static safety net:** both cores carry floating statics for 0.0.0.0/0 and 192.168.1.0/24 through the firewalls (administrative distance 200 and 201 for Primary-FW, 210 and 211 for Secondary-FW). DMZ-RT has defaults at 200 (Primary-FW) and 210 (Secondary-FW). If OSPF is lost, traffic still flows.

Firewalls, NAT and policy

Primary-FW and Secondary-FW are two independent ASA 5506 units in a routed active/active-style design. Sessions are not synchronized, so a failover reconnects users instead of preserving flows.

Interface	Zone (security level)	Primary-FW	Secondary-FW
Gi1/1	INSIDE-A (100)	10.255.1.1/29	10.255.1.2/29
Gi1/2	INSIDE-B (100)	10.255.2.1/29	10.255.2.2/29
Gi1/3	DMZ (50)	10.255.30.2/29	10.255.30.3/29

Interface	Zone (security level)	Primary-FW	Secondary-FW
Gi1/4	OUTSIDE-COMCAST (0)	65.120.100.2/29	65.120.100.3/29
Gi1/5	OUTSIDE-VERIZON (0)	71.96.50.2/29	71.96.50.3/29

- **ISP preference:** Primary-FW defaults to Comcast (distance 1) with Verizon as backup (100). Secondary-FW is reversed, preferring Verizon. Each also holds a specific route to the other ISP's service network, so both web sites stay reachable.
- **Inside to DMZ (INSIDE-IN):** ICMP, DHCP, DNS (UDP and TCP), web 80 and 443 to the two web servers, SMTP and POP3, RADIUS 1645 and 1646, NTP, TFTP, and CME ports. Everything else to the DMZ is denied. Campus and transit networks may go to any other destination.
- **DMZ to inside (DMZ-IN):** only replies from the DNS, mail, DHCP, RADIUS, NTP, TFTP and web servers, plus ICMP. Packet Tracer ASAs need these explicit reply rules for lower-to-higher security traffic.
- **Outside (OUTSIDE-IN):** Primary-FW publishes the Resume-Web and ITSupport-Web servers and allows return web and DNS traffic. Secondary-FW allows return traffic only.
- **NAT:** dynamic interface PAT per ISP for campus and DMZ traffic. Static NAT publishes 192.168.1.4 as 65.120.100.4 and 71.96.50.4, and 192.168.1.9 as 65.120.100.5 and 71.96.50.5, on Primary-FW only.
- **Inspection:** DNS, FTP, HTTP, ICMP and TFTP.
- **Management:** SSH with a local account. Packet Tracer ASAs do not support RADIUS login or failover commands.

DMZ services, DHCP, DNS and AAA

All shared services live in the DMZ at 192.168.1.0/24 behind the firewalls, so campus users reach them only through the ASA policy.

Address	Server	Role
192.168.1.1	DMZ-RT Fa0/0	Gateway for the server subnet
192.168.1.2	DHCP	Pools for every campus VLAN, reached through relay

Address	Server	Role
192.168.1.3	DNS	Internal names for the ITSupport.com servers
192.168.1.4	Resume-Web	Published web server
192.168.1.5	Email	SMTP and POP3
192.168.1.6	AAA	RADIUS on UDP 1645 for device and wireless login
192.168.1.7	NTP	Authenticated time source
192.168.1.8	TFTP	Phone and configuration files
192.168.1.9	ITSupport-Web	Published web server
192.168.1.10	DMZ-SW1	Switch management

- **AAA:** cores, access switches, DMZ-SW1 and DMZ-RT log in with `aaa authentication login default group radius local`. If the RADIUS server is down, the local account still works. Console access is local only. Each device has its own shared key, and the WLC uses the same server for WPA2-Enterprise.
- **Time:** the cores and access switches sync to 192.168.1.7 using NTP authentication.
- **Remote access:** SSH version 2 only, with ITSupport.com as the domain and an authorized-use banner on every device.

Wireless, voice and the simulated ISPs

Wireless. A WLC-3504 at 128.137.234.70 manages three lightweight APs, one on Fa0/21 of each access switch. APs get the controller address through DHCP and use FlexConnect local switching, so wireless clients land directly in their campus VLAN.

WLAN	Security	VLAN	Controller interface
Client	WPA2-Enterprise (RADIUS)	30	128.137.232.252
Executive	WPA2-Enterprise (RADIUS)	80	128.137.234.61
Guest	WPA2-PSK	100	128.137.234.124

Voice. CME-RT at 128.137.233.252 provides call control and phone provisioning for the voice VLAN. IP phones sit on VLAN 40 through the voice VLAN on each access port, with a PC daisy-chained behind each phone. CME-RT reaches VLAN 40 through its switch module.

ISPs. Comcast and Verizon are separate networks with no direct link, as in real life. Each ISP router points its default route at the firewalls and reaches the other ISP only through a third router, ISP-INTERNET-CON.

Router	Role	Key addresses
COMCAST-L3	Comcast edge and services	65.120.100.1 (circuit), 75.75.75.1 (services), 203.0.113.1 (upstream)
VERIZON-L3	Verizon edge and services	71.96.50.1 (circuit), 68.237.161.1 (services), 203.0.113.5 (upstream)
ISP-INTERNET-CON	Upstream between the two	203.0.113.2 and 203.0.113.6

Each ISP hosts a DNS server and a mock web site (Comcast 75.75.75.75 and .74, Verizon 68.237.161.14 and .15). Each circuit reaches the firewalls through its own switch and a cloud and modem pair.

Resilience and failure behavior

Every single failure below is designed to leave a working path. The table shows the intended behavior from the configuration; failover has not been exercised in the live lab.

Failure	What happens
One Po1 member link	LACP keeps the bundle up on the remaining link
BLDG1 core	BLDG2 becomes HSRP active on every VLAN, access switches forward on their Fa0/2 uplinks, and transit B stays up
BLDG2 core	BLDG1 stays active everywhere and the access switches keep their Fa0/1 uplinks
Comcast circuit	Primary-FW falls back to its Verizon default route (distance 100)

Failure	What happens
Verizon circuit	Secondary-FW falls back to its Comcast default route (distance 100)
Primary-FW	Cores and DMZ-RT use their floating static routes (distance 210 and 211) to Secondary-FW, which exits through Verizon
Secondary-FW	No user impact, because Primary-FW already carries the traffic
OSPF lost	Floating statics on the cores and DMZ-RT take over

Two limits apply. Firewall sessions are not shared, so users reconnect after a firewall failover. The published servers are NAT-ed on Primary-FW only, so they are not reachable from the Internet while it is down.

Lessons learned and known limitations

The biggest outage in this build came from one extra cable. CME-RT's switch module connects to both cores in VLAN 40, which also crosses Port-channel1, so it created a second Layer 2 path between them. The resulting loop made Po1, HSRP and OSPF flap at once. Shutting one uplink and cold-starting the lab fixed it. The final saved file has both CME-RT uplinks active again and is working, but the loop risk remains. Before reusing this design, keep CME-RT on one core or run it as a routed interface.

Other lessons from the build:

- **Two equal defaults:** when both ASAs originated a default route, the cores split traffic across them. Only Primary-FW originates one now, with floating statics as the fallback.
- **Reply rules:** Packet Tracer ASAs do not pass replies from a lower-security zone without explicit rules, so DMZ-IN and OUTSIDE-IN carry source-port reply entries.
- **ASA limits:** the 5506 model rejects failover, standby addresses, SLA tracking and RADIUS login, so redundancy is built from routing preference instead.
- **OSPF identity:** cached or duplicate router IDs cause adjacency flaps. Rebuilding OSPF with unique IDs fixes them.
- **Live state:** the saved file holds configuration, not protocol state. After a storm, saving and reopening the file is the reliable reset.

Known limitations and cleanup items:

- The static NAT for the published servers exists only on Primary-FW.
- EDGE1 and EDGE2 are unconfigured spare routers and are not part of the design.
- Failover testing, such as shutting a circuit or core, is still to be run and recorded.